Analysis of the macro-level factors which influence to suicide globally using data science and SPC tools

By
Abraham Imanol Cruz Roncal

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Dr. Wolday Abrha
Assistant professor
(Advisor)

Dr. Seong Dae Kim
Associate professor
(Committee Member)

TBD
Committee Member's Position
(Committee Member)

The University of Tennessee at Chattanooga

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EXECUTIVE SUMMARY

The present capstone project analyzes the macro factors which influence to suicide rates in an international level with the use of data science tools and applied statistics. The main purpose of this research is to identify which social, economic and health variables present the higher relation and more consistent with suicide rates across countries and throughout time.

The analysis was performed using a set of panel data combining information retrieved from several international sources, including economic indicatorids, mental health variables and data related to inequality and human capital. The methodological process included data cleaning, data transformations, multiple data base integrations and the estimation of statistic model with fixed effect for every year and country. This approach allowed to control the structural differences across countries and global shocks that affect simultaneously all the economies. The results show that the macroeconomic conditions present consistent associations with suicide rates. Additionally, variables related to substances consumption, like alcohol and drugs, also show positive effects in various specifications of the model.

In conclusion, this capstone project suggest that economic conditions and labor market conditions explain significantly suicide rates. Although mental health indicators and substance consumption also show relevant relationships, their effects interact with economic factors. Future research could amplify this analysis throughout the use of more detailed data frames or throughout dynamic models capturing the effect of long-term economic conditions and mental health results.

DEDICATION

To my family, who have always been my deepest motivation and because of their help allowing me to pursue a master's degree in the United States. To every Engineering Management professor, who supported me by teaching me insightful concepts. To life, for giving me the opportunity to learn, read, smile and enjoy.

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CHAPTER I. INTRODUCTION

Suicide is and has been one of the major causes of death around the world. In fact, it is more common among specifics groups ranging from men, youth to veterans and it is the leading cause of death among fifteen to twenty-nine years old. Despite the need of finding solutions and the efforts to improve mental health services such as psychological therapy, medicine or huge anti-suicide campaigns, suicide rates have not significantly changed throughout the last decades. As a matter of fact, suicide is the third major cause of death among young, and seventy-three percent of global suicides occur between medium and low-income countries.

1.1 Motivation

Besides economic consequences (from my point of view, the less important), suicide represents more important repercussions such as social and human. Measuring economic consequences is relatively straightforward, or at least, possible. But measuring human and social consequences is not an easy task and there is no quantitative value for human life. Thus, strategies to stop the suicide pandemic are not just important, but also an urgent issue waiting to be solved.

1.2 Background

As stated before, suicide does not affect in the same way the different types of population; instead, it attacks specific groups. For this reason, the current research intents to find those situations where each human being is more vulnerable to commit suicide. Some examples of initial independent variables are access to education and mental health institutions, GDCP, social stigma, unemployment and so on.

1.3 Problem Definition

Suicide is a problem with decades of research and interventions. Nevertheless, suicide remains as a leader cause of global health issues. Current interventions attend to treat suicide as an individual problem. The current research intents to treat suicide as a multifactorial and addresses the macro-level factors which contribute to one of the most damaging pandemics.

This study intents to find a clear understanding of suicide as a global problem and find various solutions to the moral, health priority and social issue.

1.4 Objective(s)

- To find relevant data and create a dataset with economic, social and health related factors from reliable sources with at least 65 columns within the next month.
- To identify with data science and SPC tools at least three macro-level factors which influence suicide rates within the period of two months.
- To develop an insightful project full of knowledge and suicide prevention strategies within the period of three months.

CHAPTER II. LITERATURE REVIEW

The present literature review analyzes various research papers about suicide among the last fifteen years. Although the time-scope of this literature review, and the fact that it only considers relatively new research papers, it is important to point out that suicide is a problem that has persisted for many more years or even decades.

Suicide is a major problem to public health. In 2021, 720 000 people died by suicide, which positions it as one of the leading death causes all around the world (World Health Organization, 2025). Suicide is a prevalent problem among older adults, especially in old men (De Leo, 2022), young women (Vijayakumar, 2015) and unemployed (Rajkumar, 2023), which makes this phenomena an urgent problem for the world.

Some recent studies have focused on analyzing the problem of suicide over the course of the time. Ilic and Ilic (2022) conducted a study with the purpose of studying suicide trends from 2000 to 2019. In the last year of the study, approximately 760 000 deaths by suicide were reported worldwide, and the global ASR of suicide was estimated at 9.0 / 100 000 population.

Within the study, there was consistent evidence that suicide mortality trends are affected primarily by men and suicide trends have decreased throughout time in both genders globally. However, twenty-six countries in areas as the Americas have increased their suicide mortality rates resulting in a divergence from the global trend.

Given the results, the outliers can serve as the phenomena under analysis. Or at

least, as a starting point to conduct research on the macro-factors that differentiate those countries with the rest of the world.

The difference between the countries which increased or decreased their suicide rates varied widely. Although the majority of the twenty-six countries which increased their suicide mortality rate are non-developed countries, some of them are developed countries. Thus, the results are pointing out that the macro-level factors which influence suicide are numerous and many of them are not related.

While the previous findings could make us believe that suicide is a more pronounced problem in non-developed countries, some other studies have compared the suicide rates between low-middle income countries (LMIC) and high-income countries (HIC). The results were unexpected, the suicide rates in low-middle income countries are less than the suicide rates in high-income countries (Arensman et al., 2020).

In contrast with other studies, Vijayakumar (2015) focused on suicide in women, demonstrating that despite the fact that suicide occurs more frequently in men, it is a health problem which exerts a comparable effect in women. Suicide ranks as the leading death cause in girls between fifteen and nineteen years old globally and it is more prevalent in women in countries like China.

Throughout the literature, there are different perspectives which can mislead our interpretation regarding the causes of suicide. Moreover, conclusions vary substantially depending on the data interpretation and the analytic perspective. There is a need of caution when interpreting a macro-factor and that is why conclusion should be given carefully.

Some studies have found different risk factors, like unemployment, socioeconomic status or drugs and alcohol consumption (Favril et al., 2022), while other reported correlations with macroeconomic indicators, including GDP per capita, GNI per capita, and the Gini index; although, the correlation was not consistent across all the countries and it depended on the income level of them (Lyu et al., 2025). On the other hand, some other studies have identified contributing factors as gender inequality, environmental sustainability or the nature of a nation's government (Rajkumar, 2023).

Gross domestic product (GDP) is a macro-factor that has been studied and has been associated with suicide. Blasco-Fontecilla et al. (2012) analyzed the relationship between gross domestic product (GDP) adjusted for purchasing power parity (PPP) and suicide rates. The results were the following: when the PIB per capita increased, suicide rates also increased in areas like Latin America, China, Corea, Japan and others. On the other hand, if the PIB per capita increased, suicide rates decreased in European countries, Canada, Australia, and so on. Surprisingly, in Africa a correlation was not founded.

Suicide rates and economic factors do not have the same relationship all around the world. GDP per capita does not influence suicide rates as a rule, so increasing the GDP per capita does not translate to decreasing suicide rates; instead, attention is demanded in the economic context of the country where the study is conducted.

One factor that has not been directly related to suicide rates but has been associated with happiness is inequality. Oishi et al. (2011) found in a study with data from 1972 to 2008 that Americans on average felt happier in the years with less income inequality. This

relationship varies depending on the income-group, affecting to that income groups placed below the forty percentiles, and the happiness decrease was not caused by the inequality itself; instead, it was caused by the general trust and perceived fairness of population.

The evidence clarifies that the macro-level factors do not impact directly on the object of study, but they can create some other micro-factors which have a direct influence on suicide rates. This is insightful for the current research, because giving solution to the macro-factors can induce to a top-down effect where the real causes are properly addressed.

Even though inequality influences happiness, a relationship with suicide rates is not given. Thus, an area of opportunity arises, identifying if income inequality can affect suicide rates and learn its relationship among all study variables.

Unemployment has been widely associated with suicide, particularly in men (Reeves et al., 2015) and some studies have focused on how its related specifically in adulthood (Meda et al., 2022), confirming the higher risk of committing suicide in gender and age-specific scenarios. Thus, some studies research deeper with the objective of finding how to counter-attack this relationship.

Norstrom and Gronqvist (2015) analyzed the relation between unemployment protection and unemployment suicide rates by comparing thirty countries during a sixty-year period with special emphasis on The Great Recession. The findings suggested that the better the employment protection policies, the weaker the relation between suicide and unemployment.

Across empirical studies, there is consistent evidence that unemployment

influences suicide rates. But that influence is not a rule, and its effect can increase or decrease depending on some other factors.

Considering that a variable can change its effect depending on the context. Finding a simple correlation between two variables may not be relevant. Instead, the main interest is not only finding the macro-factors which influence to suicide, but also their relationship among them.

Suicide's cost has several components, including social, human and economic; the last is the only cost that can be measured with greater or less precision. Peterson et al. (2024) set the economic cost of suicide and nonfatal self-harm to $510 billion (2020 USD) on average. This economic cost was composed of medical spending, lost work, lower quality on life and avoidable deaths. However, if a precise quantity that only considers the economic cost of suicide by itself is needed, the value is 95% of the total or $484 billion.

There is evidence that the cost of suicide is subject to diverse factors and methodologies which explain how has changed the economic cost of suicide drastically in the last years. Moreover, the cost of suicide is significantly larger that the nonfatal self-harm what changes the perception of this second one.

Although an economic cost of suicide can be set, it does not escape of interpretation because it depends on the used methodology and the decided value to life. Thus, it is important to realize the difficulty of setting a cost to suicide and to understand the complexity of this task.

Solutions with the purpose of decreasing suicide rates are hard to find because the

most obvious solutions sometimes cannot help. Andoh-Arthur and Adjorlolo (2021) compared suicide rates with macro-level mental health system indicators. The results were quite impressive; suicided rates rank higher in countries with high-income and stronger mental-health institutions.

Up to this point, it has been confirmed that suicide rates vary depending on the context and the macro-factors influence in some areas and do no not cause an increase or decrease in suicide rates by themselves; instead, they create other factors which can serve as the reason for this phenomenon.

Summarizing, a better economy, more resources or better mental health programs do not ensure lower suicide rates. This suggests a deeper problem and arises a need for researching causes of all types: economic, social, cultural, and so on, that can explain suicide.

CHAPTER III. METHODOLOGY

The current chapter explains in a justified way, our methodological strategy used to create a macro factorial country-year data frame od suicide rates and its macroeconomic, social, institutional and public health factors. The design is based on a quantitative approach with data retrieved from international sources and the gathering logic is a block integration. For instance, economic, inequality labor market, education, public health and governance.

3.1 Theoretical Framework

Our study is quantitative, longitudinal and analytic based on panel data at the country-year level. This approach is appropriate when the goal is to evaluate how suicide rates change throughout time inside each block and how they interact with each other.

Our design rationale is grounded in three factors:

- First, development indicators, government and public health used are retrieved from international data centers, particularly, based on World Development Indicators (WDI) from the World Bank, which gathers global standardized statistics used frequently in applied research.
- Second, our research question, requires a data structure which needs to conserve temporal dimension and international comparability. A country-year panel allows to document longitudinal patterns and contrast them with constant changes in social and macro conditions.

- Third, missing data in some variables (specially in the education block), gives us a path methodologically better which is a sequential block strategy, instead of creating from the beginning a balanced data frame which can reduce drastically our data size. In statistical terms, deleting data because the willing of a completely full data frame is inefficient and can frequently by skewed because of the missing data.

3.2 Data Collection

Our data collection is based on secondary data retrieves in CSV format and organized in six blocks. Each block contains indicators conceptually coherent and aligned with the explanatory framework of our research.

3.2.1 Economic factors

In the economic block macro indicators, which explain developable and stability were included.

GDP per capita, PPP (current international $). This indicator explains the GDP per capita converted to power purchasing power, which makes this indicator more suitable for our comparison of the financial well-being between countries than measures based on exchange rate.

Total unemployment (% of total labor force), defines unemployment as the proportion of labor force without a job, available and looking for a job. The International Labor Organization provides modeled estimates (ILOEST), with a specific warning, that

some data come from imputation, thus it must be interpreted carefully in national comparisons.

Inflation, consumer prices, the annual percentual variation of consumer prices index is used.

3.2.2 Inequality

The inequality block has structural measures of inclusion and distribution.

GINI index measures the deviation of income distribution with respect of perfect equality throughput the area inside the Lorenz curve and the equality line, in a scale between 0 to 100.

3.2.3 Labor Market Quality

This block explains the structural labor vulnerability.

Labor force participation, total (% of population ages 15+), is the proportion of the population which is economically active. This data is survey based and it is published annually.

Informal employment, the rate of informal employment was included as a percentage of total employment: it is composed by self-employment in informal units, familiar employment and jobs without labor rights.

3.2.4 Education

This block is designed to reflect access to opportunities.

Literacy is the percentage of population older that fifteen which can read and write with comprehension. Our comparisons between countries need to be done carefully because this data is retrieved in different ways depending on the country (surveys, tests,

etcetera).

Tertiary education is the total rate of tertiary education with respect of the population of a specific age group.

3.2.5 Health

Alcohol, recorded per capita (15+ years) consumption (in liters of pure alcohol) is used as an indicator of total APC, defined as the total consumption (registered and no registered) on average on the last three years and adjusted by tourism, expressed in liters per person older than fifteen years old by year. OMS justifies this indicator as the most precise approximation of total alcohol consumption because not registered components are included.

Depression and drug use disorders are prevalences retrieved from the Institute for Health Metrics and Evaluation throughout the GBD tools, which allows to extract estimations that can be compared by year, sex, country and age for multiple causes.

3.2.6 Government

Our intention with this block is to explain the exposition to violence and life quality.

Intentional homicides is a rate per 100,000 people and the metadata defines intentional homicide as the illicit death caused with the intention of killing or injury.

Corruption perception index is a measured elaborated by the Transparency International, which ranks countries by the corruption perception in the public sector. It reports a scale between 0 to 100 and it is retrieved from multiple sources of perception of experts and businesspeople.

3.3 Data exploration and preparation

Missing values are an explicit methodological decision because our data sources vary and do not have the same structure. Some indicators are retrieved year by year and some are not reported in the same frequency, or they are not reported (for example, inequality, puberty, employment).

Our strategy has four dimensions:

Coverage. For every indicator the coverage was calculated between the time-period required. After this study our factors were separated in high-continuity variables (inflation, unemployment, and so on) and low-continuity variables (GINI, poverty, and so on).

Specific rules depending on the indicator coverage. For high-continuity variables, an exclusion rule was applied depending on the high absence of fata. For low-continuity variables, a minimum observation quantity tule was applied as a condition to retain the country on the final data frame.

Interpolation inside each country. To avoid extensive imputation, the interpolation was limited to fill small internal empty spaces inside existent observations. This decision reduces the risk of creating false tendences without the required information. In variables with more complexity, interpolation was used with special attention and caution and was used as part of the quality control of the data.

Block factor approach instead of deletion. Instead of deleting empty rows since

the beginning (which decreases our data frame size), a master panel was preserved and an approach based on the construction of analytical specification was performed. This strategy is coherent with the literature with missing data. Deleting all missing data is inefficient and frequently skewed when dealing with missing data which depends on an indicator.

3.4 The final Data frame

The final data frame is a composition of a key decision: using suicide rates data as the basis and merge the other variables with a strategy type left join by country and year. The procedure assures that the result preserves the complete data from the dependent variable and that the missing data was concentrated in the independent variables when a specific source does not retrieve values for a specific year or country.

The reason of this methodological decision has its foundation in our research question, and that is why if an inner join was used instead, our final data frame would be reduced to the size of the less complete variable, which affects the research validity skewing our data to those countries with more statistical capacity.

After creating the final data frame, an integrity control was executed. Checking duplicated values, comparison of available years and the calculation of the percentage of missing values by macro factor. This diagnosis serves to define the block strategy in the analysis, first variables with high-continuity and then a sequential incorporation of variables with less coverage, documenting changes in the data frame size and countries

used inside the study.

3.5 Software and tools

Our study was performed in Jupiter notebook using Python and standard libraries for data science such as pandas and NumPy. Reproducible data cleaning and transformation procedures were employed for retrieving, reading, transforming wide to long, depuration of country names, duplicates treatment, and generation of merges. Additionally, an organization in folders to guaranty the repeatability and tradability of each macro factor was implemented.

To ensure repeatability each block keeps the same workflow: importation, data cleaning, missing values treatment, exportation of the data frame and integration to the final data frame. What is looking with this approach is that the reader and evaluator can replicate the results with precision.

3.6 Limitations

There are some limitations inherent to the data itself and the integration strategy.

Several indicators have international restrictions. For example, urbanization depends in the national definition, and the interpretation need to be performed carefully. In the same manner, literacy can be measured throughout test or surveys, with bid differences between countries.

Some macro factors are modeled by ILO which incorporates estimation and imputation to complete the coverage, which increases the uncertainty and limits the use for a strict study.

Regarding the study decision, we established explicit limits and the analysis is restricted by a country-year level and the final data frame is based on the availability of the dependent variable. A block study approach is used to incorporate variables with less coverage, avoiding that a single incomplete indicator affects the entire study. These decisions are coherent with the research objective: identify which macro factors influence more consistently with the variation of suicide rates.

The final data frame is presented in Table 1. The final structure of our data frame after cleaning, filtering, dealing with missing data, and interpolation is observed. Every row represents a country-year observation, and every column is a macro factor which con influence suicide.

Table 1: Final data frame of macro factors which influence to suicide

country	**year**	**suicide_rates**	**gdp**	**inflation**	**unemp**	**labor**
Cabo Verde	2011	18.243333	6384.228441	4.473883	10.956	59.129
Thailand	2002	11.333333	8148.437029	0.697309	1.82	73.27
Guatemala	2002	12.233333	4969.257449	8.132631	2.254	66.953
Nepal	2015	12.373333	2957.305591	7.868909	10.643	40.224
Bhutan	2012	5.723333	8577.407319	10.919657	2.05	64.4
China	2004	15.103333	4504.929372	3.824637	4.49	72.927
Barbados	2012	3.533333	19605.01693	4.533348	11.59	67.569
Portugal	2014	9.556667	28764.5255	-0.278153	13.88	58.777

1 Nan values indicate missing data due to data unavailability in the original data frames and depending on the interpolation rules these empty values were not treated.

Interpolation was restricted to internal gaps.

The complete data set contains several more macro factors not displayed due to the page size.

CHAPTER IV. ANALYSIS

4.1 First block

In the analysis of the first block the influence of factors related to mental health and drugs consumption to suicide was evaluated using a panel data model with a year-country structure. Initially, logarithmic transformations were applied to the depression and drug consumption variables due to their high asymmetry with the purpose of improving the linearity and the model stability. Right after two different models were deployed: a simple OLS regression (pooled) and a model with fixed effects for country-year.

The OLS model simply captured the structural differences between countries, showing significant associations bit significantly skewed, due to unobservable characteristics that are constant throughput time (culture, health system, statistical gathering quality, and so on). After incorporating fixed effects for every country-year, the analysis evaluated variations inside the same country throughout time, controlling simultaneously global annual shocks.

In the fixed effects model, both depression rate and drug consumption showed positive effect and statistically significant to suicide. This means that inside a country, increases in these variables throughout time are associated with increases to suicide rates. On the other hand, alcohol consumption lost statistical significance once the structure differences were controlled, what suggests that the effect of alcohol consumption in the

simple model answered mainly to the constant differences between countries and not the dynamic variations inside them.

Additionally, we evaluated the possible multicollinearity between variables. Although the initial VIF (Variance Inflation Factor) suggested high values, the correlation analysis showed a moderate relationship between depression and drug consumption (0.24), discarding severe collinearity between them. The apparent VIF inflation could be explained due to the limited temporal variation inside each country, a common characteristic in macrostructural indicators. This restriction does not invalidate the model, but it implies that the estimated effects are based on relatively small changes throughput time.

Summarizing, the first block demonstrates that mental health conditions and substance use exhibit a solid association with suicide rates when the internal dynamic between each country is analyzed.

Figure 1 presents a comparison between the coefficients estimated for the OLS simple model and the fixed effect model for the three first variables related to mental health and substance consumption. The objective of this figure is to visualize how estimations change when the model changes from a simple OLS to a more rigorous study which controls fixed effect for country and year.

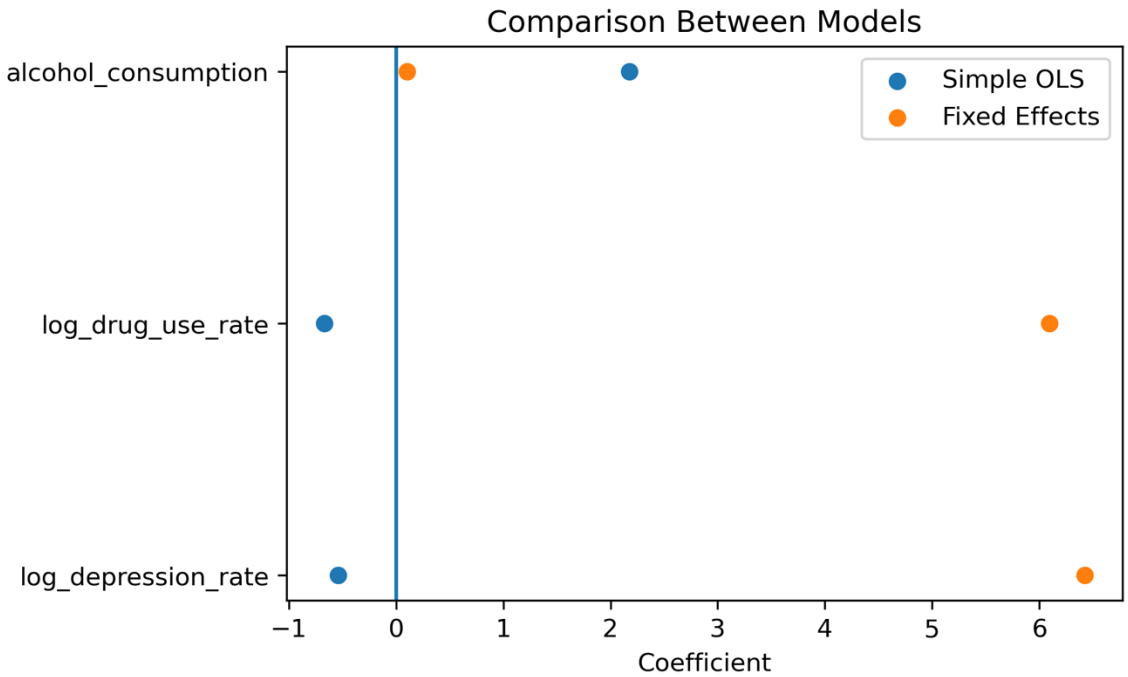

Figure 1: Comparison of Coefficients Between Simple OLS and Fixed Effects Models

4.1.1 OLS model

The table 2 presents the result for the Model 1, which corresponds to the baseline empirical analysis. This model evaluates the correlation between three variables related to mental health and the substance consumption: depression rate, drug consumption and alcohol consumption per capita, with suicide rates.

The model was estimated using ordinary least squared (OLS) with a total of 4004 observations. The determination coefficient is 0.080 and indicates that approximately 8% of the observed variation in suicide rates can be explained with the included variables in

this first specification. This relatively small value is expected in preliminary models because they do not incorporate structural factors or macroeconomic.

The result show that alcohol consumption presents a positive coefficient and is highly significant, what suggest that increases in alcohol consumption are associated with higher suicide rates. These findings are consistent with literature which documents that the use of substances es a major factor of auto destructive behavior.

Conversely, depression rates and drug consumption present negative coefficients for this specification. Although both coefficients are statistically significant, their interpolation should be performed with caution, because the model does not control structural differences between countries.

In consequence, this model is an initial approximation which identifies simple associations between mental health factors and suicide. The next model incorporates macroeconomic factors and fixed effects per country and year to isolate with more precision the determinants of these phenomena.

Table 2: OLS Regression Results

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          suicide_rates   R-squared:                       0.080
Model:                            OLS   Adj. R-squared:                  0.079
Method:                 Least Squares   F-statistic:                     115.3
Date:                Wed, 04 Mar 2026   Prob (F-statistic):           1.34e-71
Time:                        03:07:02   Log-Likelihood:                -13267.
No. Observations:                4004   AIC:                         2.654e+04
Df Residuals:                    4000   BIC:                         2.657e+04
Df Model:                           3
Covariance Type:            nonrobust
==========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                   10.3310      0.105     98.265      0.000      10.125      10.537
log_depression_rate     -0.5403      0.109     -4.960      0.000      -0.754      -0.327
log_drug_use_rate       -0.6658      0.121     -5.518      0.000      -0.902      -0.429
alcohol_consumption      2.1768      0.120     18.215      0.000       1.943       2.411
==============================================================================
Omnibus:                     1042.627   Durbin-Watson:                   0.580
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2628.722
Skew:                           1.414   Prob(JB):                         0.00
Kurtosis:                       5.786   Cond. No.                         1.75
==============================================================================
```

1 This baseline estimates the relationship between mental health macro factors, alcohol and drug consumption, and suicide rates across countries.

4.1.2 Fixed Effects Regression with country and year controls

The Table 3 represents the results for the second model, a regression with fixed effects which incorporates controls for country and year with the purpose of capturing the not observable heterogeneity between countries and the common global shocks throughout time.

The main difference with the base model is that this specification introduces variable indicators for each country and for each year, what allows to isolate the effect of explanatory variables to suicide rates considering just the internal variation inside each country throughout time. In other words, the model compares how suicide rates change

inside a country when the explanatory variables change, deleting the structural differences between countries.

The model presents a determination coefficient of 0.924, what indicates that the explicit specification explains a great proportion of the observable variability in suicide rates once the structural differences between countries and the temporary global effects are controlled.

In this specification, alcohol consumption maintains a high positive coefficient and statistically significative. On the other hand, depression and drug consumption show positive coefficients and statistically significant in this model.

Table 3: Fixed Effects Regression Results with Country and Year Controls

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          suicide_rates   R-squared:                       0.924
Model:                            OLS   Adj. R-squared:                  0.920
Method:                 Least Squares   F-statistic:                     224.8
Date:                Wed, 04 Mar 2026   Prob (F-statistic):               0.00
Time:                        03:33:24   Log-Likelihood:                -8277.5
No. Observations:                4004   AIC:                         1.697e+04
Df Residuals:                    3798   BIC:                         1.826e+04
Df Model:                         205
Covariance Type:            nonrobust
==========================================================================================
```

1 This fixed effect estimates the relationship between factors while controlling country and year specific effects.

4.2 Second block

The second block increased the first model of the first block incorporation a set of economic variables, with the objective of evaluating if the observed associations between suicided and the mental health macro factors keep having the same weight after

implementing macroeconomic conditions into the model. Thus, a country-year panel was constructed keeping suicide rates as the independent variable. With the purpose of facilitating the interpretation and reducing the observed asymmetry, logarithmic transformations were performed for those variables with a high skewed distribution.

While variables as GDP per capita, depression rate, drug prevalence were affected by the logarithmic transformation, some others as alcohol consumption, unemployment and inflation were included in their original scale. Additionally, to compare relative quantities between predictors, explicative variables were standardized (mean cero and standard deviation one), expecting that the estimated coefficients could represent the expected change in our independent variable after an increase of a standard deviation in each predictor.

An analytical subset was defined with the elimination of a whole row to guaranty comparability between variables and avoid that the model could estimate inconsistent estimation of information. These criteria reduced our sample size to 3,456 observations, 159 countries and 22 years which represents a wide enough panel structure but with the drawback of not been balanced, due to countries with missing information for the 22 years. The reduction of the final data frame is expected in macro international analysis due to data which is not identically distributed between indicators. For the final model, the presence of all variables is required, so a stricter intersection is obtained.

The estimation of the second block was performed throughout a regression model with fixed effect for country and year. This specification controls unobservable

heterogeneity that is constant in the time inside each country (culture, economic structure, mental health systems, and so on), and, simultaneously, for common shocks in a certain year affecting multiple countries (For example. International, crisis or global tendences). In consequence, the coefficients of the independent variables are interpreted as relationships within country.

Within country relationships are changes in suicide rates associated in the independent variables variations inside each country throughput time, keeping the structural effects and common variation of every year constant.

Moreover, robust clustered standard errors were implemented for country to correct possible future problems due to heteroscedasticity and the serial correlation inside each country. These adjustments enhance the inferential validity of both the p-values and the confidence intervals.

Under these specifications, our model allowed to identify consistent patterns: GDP per capita showed a negative effect and statistically significant, what suggests that increases in the economic level inside a country is associated with reduction in suicide rates; unemployment show a positive effect and statistically significant, coherent with previous research, which states that labor market deterioration increase the risk; and depression kept a positive and significant effect even after adding economic macro factors into the model.

Figure 2 shows the evolution in the estimated coefficients for the principal variables related to mental health and drugs and alcohol consumption, comparing the Model 1 with

the Model 2. Whereas Model 1 includes only mental health and consumption variables, the Model 2 incorporates macroeconomic variables, what allows to evaluate if the initial observed effects remain once the macroeconomic factors are considered.

Depression and substances consumption association decrease considerably from Model 1 to Model 2. This change suggests that the estimated effect in model 1 was being captures indirectly by the economic conditions which also influence to suicide rates. After introducing macroeconomic variables, our model accomplishes to separate with higher precision the effect associated with mental health.

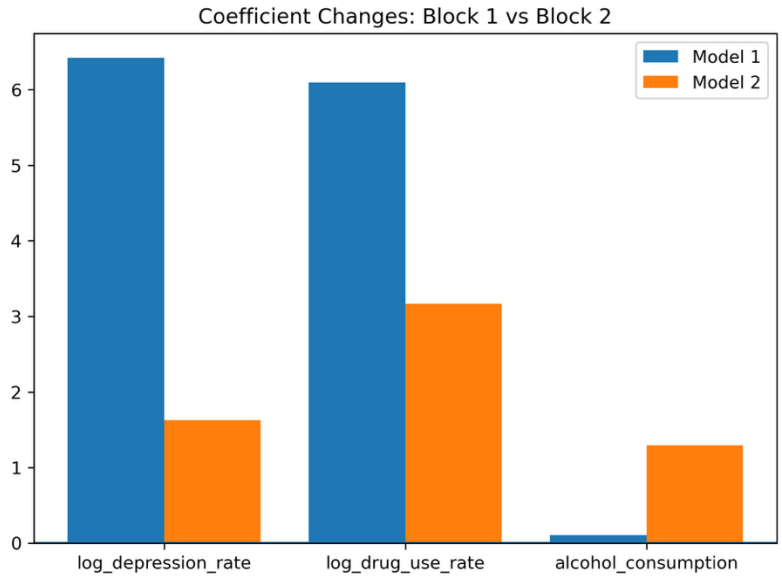

Figure 2: Coefficient Changes: Block 1 vs Block 2

Drug and alcohol consumption kept positive effects, but with evidence statistically less significant (marginal significance), what implies more uncertainty in their independent influence in suicide rates. Finally, inflation did not show any effect statistically different than cero, suggesting that, under country-year countries, their variables do not have correlation and do not influence directly to suicide rates or our analyzed sample.

Synthetizing, the second block has two different and important methodological functions. First, evaluating the robustness of our first block (mental health) when macro-economic variables are added; and second, stablish a comparative framework of relative magnitudes under standardization and fixed effects. Our result suggest that economic dimension contributes to the explanatory explicative power and, even that, depression conserves a robust positive association with suicide rates inside each country throughout time.

Table 4 shows the result of the regression model incorporating simultaneously mental health variables, substance consumption and macroeconomic factors. The table reports estimated coefficients, p-values and confidence intervals in 95%, what allows to evaluate the magnitude and statistic significance of each variable related to suicide rates.

Table 4: Regression Results with Macroeconomic Controls (Model 2)

	Coefficient	p-value	CI Lower	CI Upper
log_depression_rate	1.6251	0.007	0.4439	2.8063
log_drug_use_rate	3.1641	0.0574	-0.0999	6.4281
alcohol_consumption	1.2973	0.0721	-0.1164	2.7111
log_gdp	-3.2955	0.0007	-5.2001	-1.3908
unemp	0.872	0.0001	0.434	1.3099
inflation	0.0344	0.6291	-0.1052	0.1741

1 The results of the second fixed effects regression model and its, p-value, and confidence intervals.

The result of the second block confirms that both mental health factors and economic conditions influence significantly to to suicide rates and that part of the effect observed in the first model was caused by the omission of relevant variables. These findings justify working with blocks and adding new ones to analyze if these factors modify those correlations.

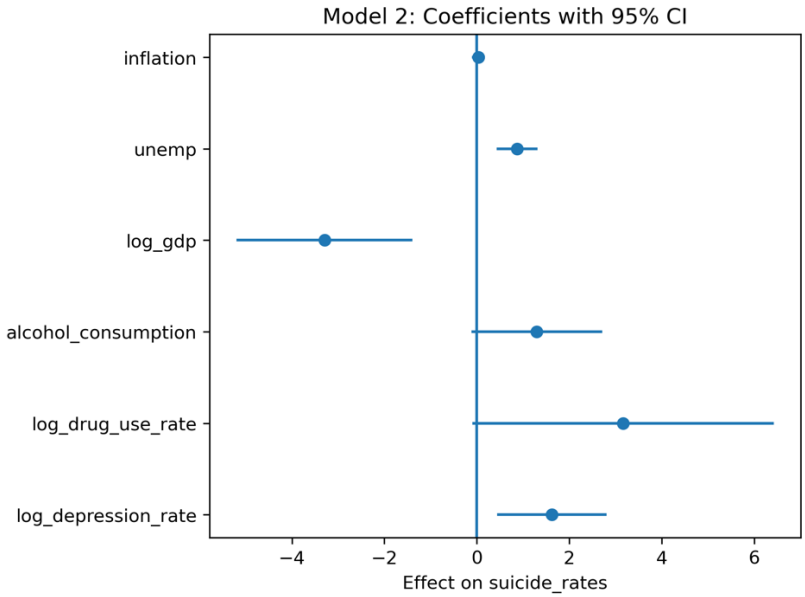

Figure 3: Coefficient of Model 2 with 95% CI

4.3 Third block

The second block allowed our research to identify the relevance of macroeconomic factors and mental health factors when explaining suicide rates. These results do not capture completely all the structural dimension related to the structures resource distribution and the access to opportunities. Particularly, economic development does not necessarily reflect how income is distributed inside a country and the effective grade of education access and human capital.

For this reason, the next Bloque analyses and incorporates variables associates with inequality, like the GINI coefficient, poverty levels and literacy, and so on. These variables allow to evaluate of the socioeconomic disparities, and the education opportunities have an additional effect and influence suicide rates.

Methodologically, the structure of fixed effects for country and year is kept, and the use of clustered standard error in a country level, with the purpose of assuring comparative consistence between blocks and preserve inferential robustness. The strategy of staggered blocks allows to identify how the progressive inclusion of structural dimensions modify the magnitude and significance of previous estimated coefficients.

Due to data availability and statistical stability of our analysis. The third block was divided in two specifications (Model 3A and model 3B).

First a lot of the structural variables included in the last block, particularly the relate to human capital (tertiary education and literacy) present limited coverage in global level.

This means that there is not data for all countries and there is not available data for all years. If all variables were included simultaneously in a unique model, our data frame size would be reduced drastically, what could affect the precision of the estimation and d=the capacity of generating results.

In Model3A, variables related to economic inequality were incorporated (GINI index and poverty rate), maintaining a sample size relatively wide of countries and observations. This allowed to evaluate if such factors exert an additional effect to suicide rates, beyond the macroeconomic variables and mental health variables considered in previous models.

Table 5 shows the result corresponding to Model 3A, in which the latest specification is amplified by the. Incorporation of variables related to inequality and poverty. This block's objective is to evaluate if the distributive factor exerts an additional influence in suicide rates once mental health, substance consumption and macroeconomic conditions are controlled.

Table 4: Regression Results with Inequality and Poverty (Model 3A)

	Coefficient	p-value
log_depression_rate	2.3162	0.0036
log_drug_use_rate	2.5457	0.1674
alcohol_consumption	2.1384	0.0171
log_gdp	-6.673	0.0002
unemp	0.7645	0.0005
inflation	0.0455	0.6467
gini	0.0746	0.8383
poverty	-0.825	0.0626

1 Estimated coefficients and p-values for model 3A

In the Model 3B, variables associated to human capital were included in the study, as tertiary education and literacy. Due to the lack of data availability of these indicators, the use of these variables implied a considerable reduction in the number of countries and analyzed observations. However, these specifications explore if the educative factors and access to opportunities modify the observed effects in previous models.

Table 4: Regression Results with Literacy and Tertiary Education (Model 3B)

	Coefficient	p-value
log_depression_rate	0.7977	0.5509
log_drug_use_rate	5.1058	0.0212
alcohol_consumption	1.5911	0.0572
log_gdp	-3.1065	0.0451
unemp	1.1114	0.0031
inflation	0.1005	0.3767
gini	0.0502	0.9139
poverty	0.3033	0.5798
tertiaryeduc	0.2786	0.5849
literacy	-1.1867	0.228

1 Estimated coefficients and p-values for model 3B

The separation of the last block in two models, accomplishes two methodological objectives. First, to preserve the most sample size, if possible, when variables with good international coverage are analyzed. And second, to evaluate if the effect of variables with less availability without compromise the stability of the estimations of the principal model. With this approach the sequential strategy allows to compare result between specifications and evaluate the robustness of the estimated effects.

Figure number 4 compares, variable by variable, how estimated coefficients change from Model 3A to Model 3B. Each point is the coefficient, and the horizontal bar is the confidence interval.

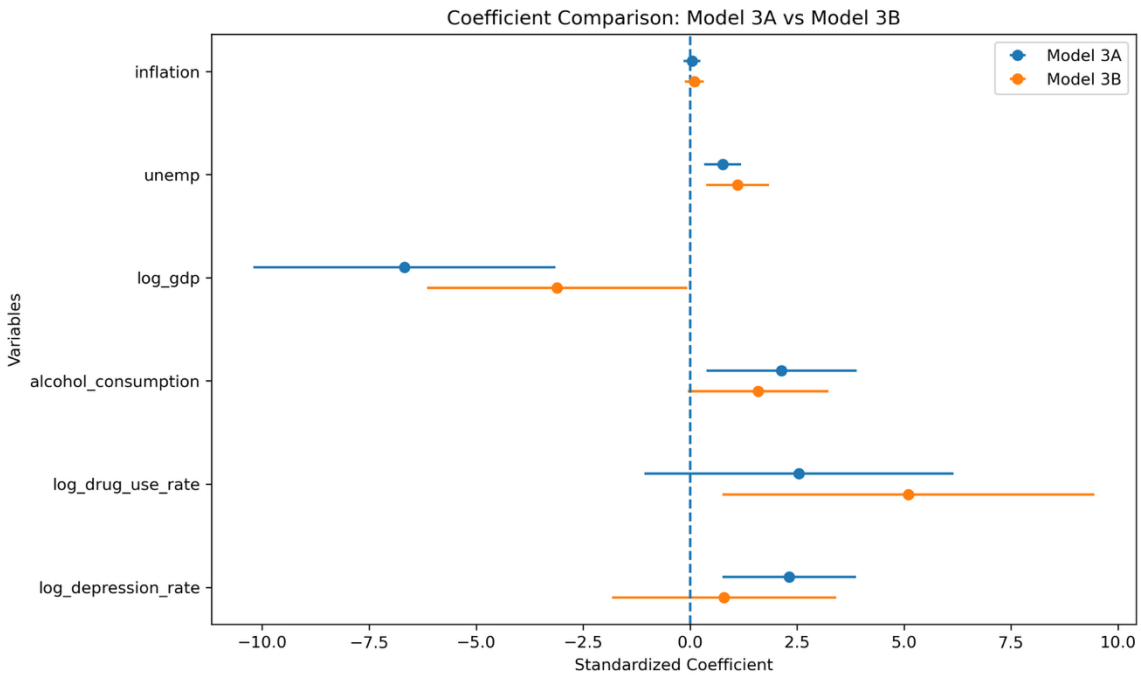

Figure 4: Coefficient comparison: Model 3A vs Model 3B (with 95% confidence intervals)

In the comparative analysis of the second block and the third block, we found a structural pattern concise and consistent. PIB per capita and unemployment exhibit robust and stable effects and in all specifications. A higher economic development is associated systematically with lower suicide rates, whereas an increase of unemployment increases suicide rates. These effects remain statistically significant even after incorporating inequality and human capital factors, what suggests that the macro-economic conditions

and the labor market are determinant if the suicide phenomena.

In second place, the inequality of income, measured by the GINI index, did not present dynamic effects statistically significant inside countries once fixed effect are incorporated. The result indicates that the structural differences between countries in terms of inequality would be more relevant than the short-term fluctuations inside each country.

Third, poverty shows instability between specifications, changing sign and losing significance when the data size is reduced. This behavior suggests that poverty effect is sensible to the sample compositions and it is possibly correlated with other macroeconomic variables, which can generate problems of multicollinearity or capture structural effects.

Mental health variables present a heterogeneous effect. Depression results robust in big sample sizes but it losses significance in restricted data sizes. On the other hand, drug consumption acquires significance in restricted data sized, what suggests the existence of heterogeneity between groups of countries or differences in the quality of data acquisition and data coverage.

Finally, human capital reduces drastically the data size, showing that global availability is a limitation in international panel analysis. The empirical evidence suggests that the economic structure and the market conditions determine more stable suicide rates than the inequality short term variables.

CHAPTER V. CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

The present research analyzed the relationship between mental health indicators, substance consumption, macroeconomic conditions, and human capital variables in the explanation of suicide rates across countries and throughout time. To accomplish this task, a data frame was constructed which integrated all information retrieved from several international sources combining economic, social and public variables.

The main purpose of this research was identifying which factors have a stronger and consistent influence with suicide rates and determine if the economies conditions, mental health indicators and other distributive variables. Contribute in the same significant manner for the explanation of the differences observed across countries and throughput years.

The empiric strategy follows a stepwise focus in 3 different data blocks. In the first phase, the analysis centered in variables related to mental health and alcohol and drug consumption. After that, macroeconomic variables were included like PIB per capita, unemployment and inflation. Finally, the model amplified to include variables related to inequality, poverty, tertiary education and literacy. In the same manner, fixed effect per country and year were included with the purpose of controlling non observable structural characteristics and global shocks which can influence simultaneously across all countries.

The analysis results show various consistent patterns. In the first place,

macroeconomic conditions perform an important part of the explanation of suicide rates. Throughput different model specifications, PIB per capita presented a negative correlation and statistically significant with suicide rates. This result suggests that the higher development levels, the lower suicide rates. The phenomena can be explained by better life conditions, and more health access.

On the contrary, unemployment showed a positive relationship and statistically significant with suicide rates. This finding show that increases in unemployment inside countries are associates with increases with suicide rates, what is consistent with the existent literature.

The variables related with substance consumptions also presents relevant associations with suicide rates. Alcohol consumption shows a general positive relation with suicide in various model specifications, but its significance varies depended on the model. In the same manner, drug consumption presents a positive correlation in some models.

Depression shows a more complex behavior. In bigger data samples, depression rates present a positive and significant relation with suicide rates. Nevertheless, when more variables are incorporated and the sample size is reduced due to data availability limitations the statistical significance decreases. This result suggests that some of the relation between suicide and depression could be related to other social and economic factors included in the model.

Inequality and poverty did not provide statistic evidence about the short-term

income distribution and its relationship with suicide rates. The GINI coefficient does not result statistically significant in estimation model and poverty present instable effects, suggesting that the structural differences across countries in terms of inequality could be more relevant that the temporary fluctuation captures in the study.

Tertiary education and literacy do not show statistically significant evidence in the final model. This result could be explained by the limitation in data availability for these variables, what decreases the data size and the precision of the estimations.

Summarizing, the results of this study show that the macroeconomic conditions and the market labor conditions are consistent factor determining suicide rates across countries. Although mental health and substance consumption also have relevant associations, their effects, their effect apparently interact with the economic context where they are developed.

5.2 Recommendations and Future Studies

With the basis of the results of this research, it is possible to propose some recommendations for the design of public politics and academic research lines.

The responsible for public politics should recognize the strong relationship between economic conditions and suicide rates. Unemployment presents a strong relationship consistent with suicide rates, so politics with the purpose of strengthen labor market and promote economic stability could contribute indirectly to reduce suicide risk.

Drug consumption is an important issue because it presents positive correlation with suicide. Thus, public health campaigns with the purpose of preventing substances

consumption and the expansions of treatment programs could help to reduce suicide risks.

For future research, one of the hardest tasks is related to the availability and quality of international data. Several of relevant variables present huge quantities of missing values. Future research could beneficiate of improvement in the systems of data collection and the integration od new global data bases.

Suicide determinants could vary depending cultural differences. Focused research in specific regions could improve a better comprehension of the macro factors which influence suicide inside specific contexts.

In conclusion, the results if this research suggest that suicide prevention require a multidisciplinary focus which combines economic politics, public health strategies and social help. Comprehending the interaction between macro factors is fundamental to design effective interventions with the purpose of reduce suicide risks.

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APPENDIX A: JUPYTER NOTEBOOK, PYTHON CODE AND FULL ANALYSIS

JUPYTER NOTEBOOK, PYTHON CODE AND FULL ANALYSIS

The full process of data preparation, data bases integration, logarithmic transformations and econometric model estimations is documented in a jupyer notebook. With the purpose of promote transparency and repeatability, the code used in this capstone project is public and can be found in the following repository:

GitHub: https://github.com/imanolcruzroncal